

Lab 2 - Linguistics Data, Stat 215A, Fall 2026

October 3, 2026

1 Introduction

Language is a deeply human tool that not only is used to communicate, but also shape thoughts, ideas, and emotions. As a tool, it can also change with usage, and the variation of language can reflect important historical, social and geographical factors of a society (Nerbonne and Kretzschmar 2003). Dialectometry is a field of linguistic studies that specifically measures dialect differences, or "linguistic differences whose distribution is determined primarily by geography" (Nerbonne and Kretzschmar 2003). Given that these are differences within the same language, a main question that arises is whether dialects differ based on distinct dialect areas or more as a gradual variation across regions. In this report, we will try to answer this question by looking at the results from an online dialect survey. The principle idea is that by considering multiple features of responses from people, patterns of dialect differences will be more detectable than individual features that may be noisy or only relate weakly to geography.

Throughout this report, I will try to answer the following question: Do people group into distinct lexical regions, or do they vary along a continuum? In order to answer this, I cleaned the dialect survey dataset, ran initial exploratory analysis on a couple of questions, evaluated different dimension reduction techniques, conducted clustering analysis, and discuss the stability of these approaches.

2 The Data

The data we will be using are the 47,471 responses to Bert Vaux's online Dialect Survey. We will be looking specifically at the responses for a subset of the questions that focus on lexical differences, such as asking about which word they use for specific concepts. The respondents also reported their city, state, and zip code. The zip codes were used by former GSIs of the course and were converted into latitude and longitude coordinates.

2.1 Data Cleaning

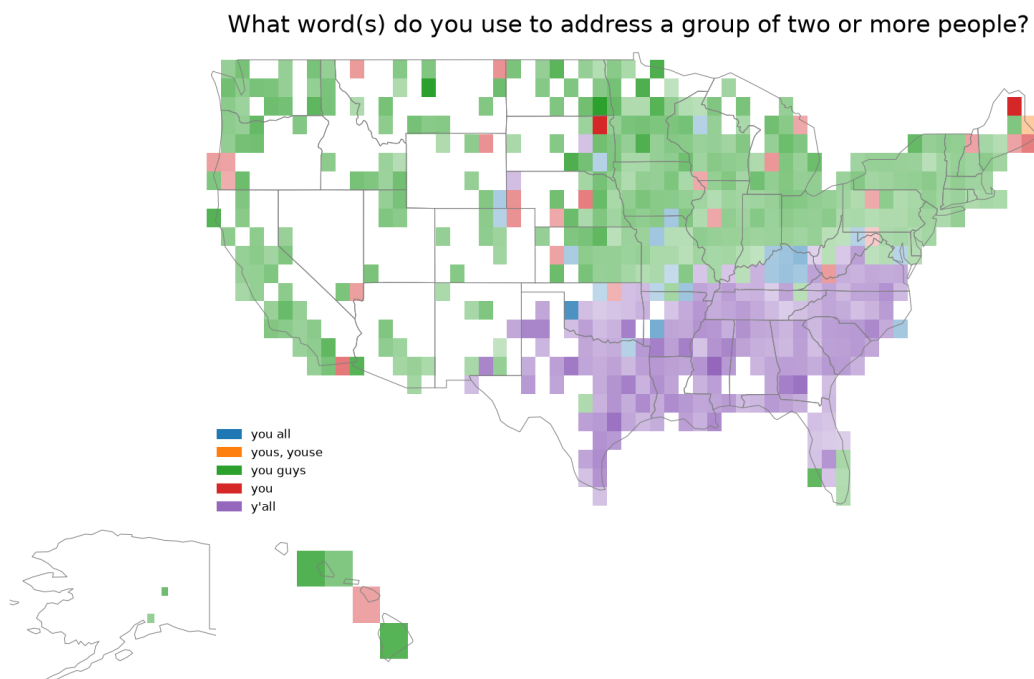
Of the 47,471 responses, there were 1,020 responses that did not have latitude and longitude information. These missing values seem to be mostly concentrated in a few states (MA, CA, VA, FL) and some cities (Newton, Boston, Lexington, etc.). This could be due to the zip code not being properly formatted, since the city and state information may not have been used to derive the coordinates. These 1,020 responses have been dropped from the following analyses in order to use the coordinates for mapping the results. This led to these states and cities being disproportionately excluded for analysis. A possible alternative data cleaning choice would have been to fill in the information using the city and state responses.

Next, I looked at non-response rate for each person. 1,040 people did not answer any of the 67 questions in the dataset we are analyzing. These people have been dropped from subsequent analyses. Partial blanks (i.e. if at least one question has been answered) are kept in the dataset. This resulted in a final dataset of 45,436 respondents.

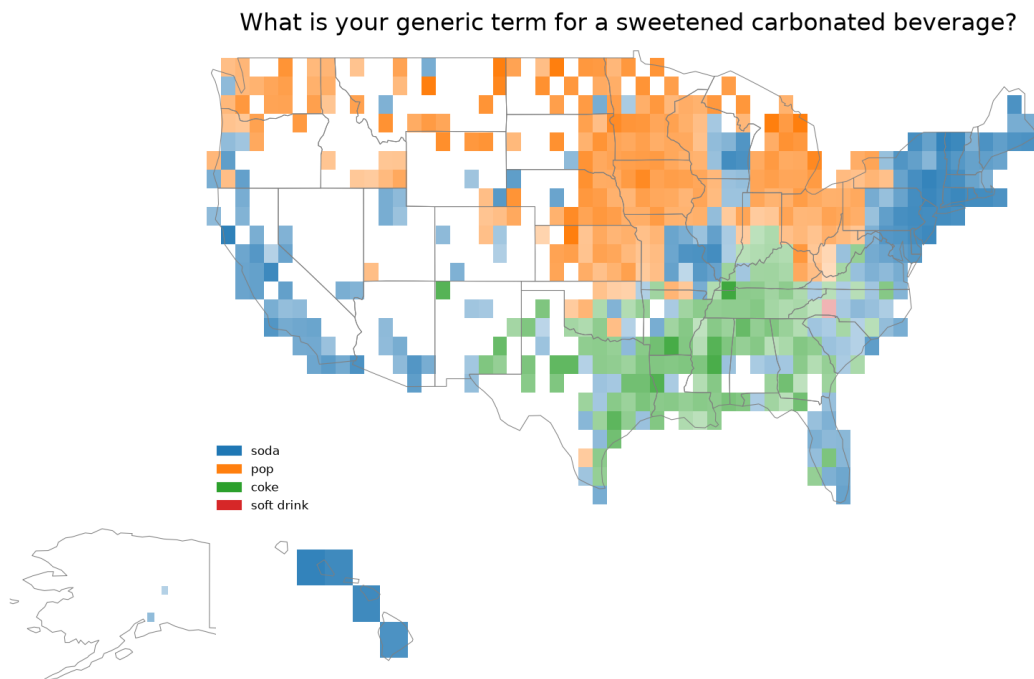
2.2 Exploratory Data Analysis

First I focused on two questions from the survey that are well-known regional words. Figure 1 maps the top answers from each 1° latitude and longitude bin. The stronger each color is, the more dominant the top answer is within that bin.

Figure 1a shows a general split between the South (y'all) and North + West (you guys). Figure 1b shows a general split across the East and West coasts (soda), the Midwest (pop), and the South (coke). However,



(a) Q050: Group of people



(b) Q105: Sweetened carbonated beverage

Figure 1: Most common answer in each $1^\circ \times 1^\circ$ latitude, longitude bin, among respondents who answered the question. Bins with fewer than 5 respondents were dropped. Opacity is proportional to the percentage of the bin giving the top answer (minimum 15%).

none of these are a perfectly clean split with mixed bins, suggesting that the single features do not have enough signal to show clear lexical regions.

If we were just to use the answers for one of the questions, could we predict the answer for the other question? I used a random half of the data to learn the most common Q105 answer for each Q050 answer. Then, on the held-out half of the data, I used the prediction rule learned from the the Q050 answer to test how often this was correct. I also used a prediction rule of always guessing the most common answer (soda) to see whether having the Q050 answer was a useful predictor.

Q050 answer	<i>n</i> (test)	Predicted Q105	% correct (lookup)	% correct (always soda)
you guys	9,340	soda	49.1	49.1
you	5,278	soda	55.6	55.6
y'all	3,945	coke	45.0	31.9
you all	2,900	soda	47.6	47.6
other	717	soda	48.1	48.1
yous, youse	156	soda	67.3	67.3
yins	63	pop	79.4	19.0
you 'uns	45	pop	62.2	11.1
you lot	34	soda	44.1	44.1
All	22,478		49.9	47.3

Table 1: Predicting Q105 from Q050 on the held-out half. For each Q050 answer, the prediction is the most common Q105 answer among people in the training half who gave that Q050 answer.

As shown in Table 1, the overall lookup rule is 49.9% correct whereas the baseline is 47.3% correct, suggesting that there is some predictive improvement. When we look at the specific answer pairs, this seems to be driven by the fact that the Q050 answer "y'all" predicts Q105 answer "coke". This could be due to a regional relationship between these dialect choices. However, the small boost in prediction also points to the fact that one question does not tell us much about the answer to another at an individual level.

3 Dimension Reduction

Our efforts to investigate relationships between two questions has led to the conclusion that it might be more informative to consider multiple features at once to try to capture region-based lexical differences. So next, we tried different methods of dimension reduction.

After data cleaning, I one-hot encoded the responses such that the answer codes become labels and each question gets one binary column coded per answer. To illustrate this process better, I will reiterate the example from the lab2 instructions. Suppose the questionnaire had only two questions, the first with three answer choices (A–C) and the second with four (A–D). If one participant answered A and D and another answered B and D, the encoded data is the 2×7 matrix in Table 2. Each row has exactly one 1 per answered question, and a question left blank becomes a block of all zeros.

	Question 1			Question 2			
	A	B	C	A	B	C	D
Participant 1 (A, D)	1	0	0	0	0	0	1
Participant 2 (B, D)	0	1	0	0	0	0	1

Table 2: One-hot encoding of two answers to a two-question survey. Columns are grouped by question; each answered question contributes a single 1.

Applied to all 67 questions in our dataset, we got a $45,436 \times 468$ matrix.

First, I ran PCA on this one-hot encoded matrix. This was necessary because the responses in the raw dataset are nominal (e.g. soda = 1, pop = 2, coke = 3) and running PCA on the raw dataset could lead to the interpretation of the raw codes as a scale difference (e.g. coke would be 3 times soda). I've centered each of the columns so that resulting PCs describe the direction of deviation from the average answer. I did

not scale the data because the questions have equal weight and dividing by the standard deviation would actually inflate rare answers.

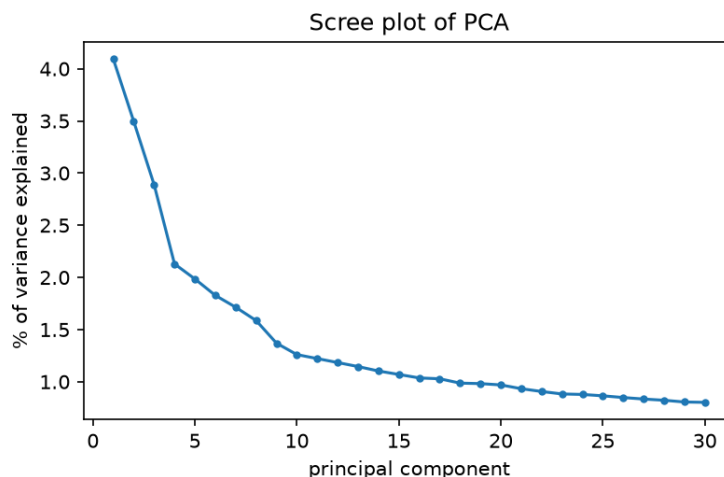


Figure 2: Percentage of variance explained by each of the first 30 principal components.

Figure 2 shows how much variance of the data is explained by each PC. There isn't a clear elbow, but the initial steep slope changes a bit at the 4th PC.

Figure 3 shows the mean PC score for each of the 4 PCs. PC1 shows a clear positive scores concentrated in the Northeast regions. The specific answers with positive loading on this PC include words like "sneakers", "soda", and "sunshower". PC2 shows a North vs. South gradient and the scores seem to gradually change through the middle. Positive loadings on PC2 include phrases like "kitty-corner", "drinking fountain", and "firefly" whereas negative loadings are "catty-corner", "water fountain", and "y'all". PC3 seems to not show a clear regional grouping and the colors are dispersed throughout the map.

I tried running the PCA analysis with different centering and scaling choices as well to illustrate the effects of these choices. Not centering the data first results in PC1 reflecting the average answer and the later PCs are the same as the centered PCs. Standardizing the data results in PCs 1–3 that look the same as the centered ones, but PC4 weighs the rare answers much more, such that it dominates.

We can also run NMF since our data matrix is the response to the questions and thus is nonnegative. Figure 4 shows that the error decreases steadily with k and again has no elbows. I chose $k=4$ to match the 4 PCs. NMF separates out different parts of the answer that would make up each individual's answer as a weighted sum. The four profiles shown in figure 5 illustrate that these answer profiles show some regional hot spots of the Upper Midwest, Northeast, South, and West, but mostly have gradual edges, rather than clear distinctions. Note that these solutions could change with the k value and the random start.

With two different dimension reduction approaches, similar broad regions such as Northeast, Midwest, and South were recovered, suggesting that the dialect data seems to vary based on geographical factors.

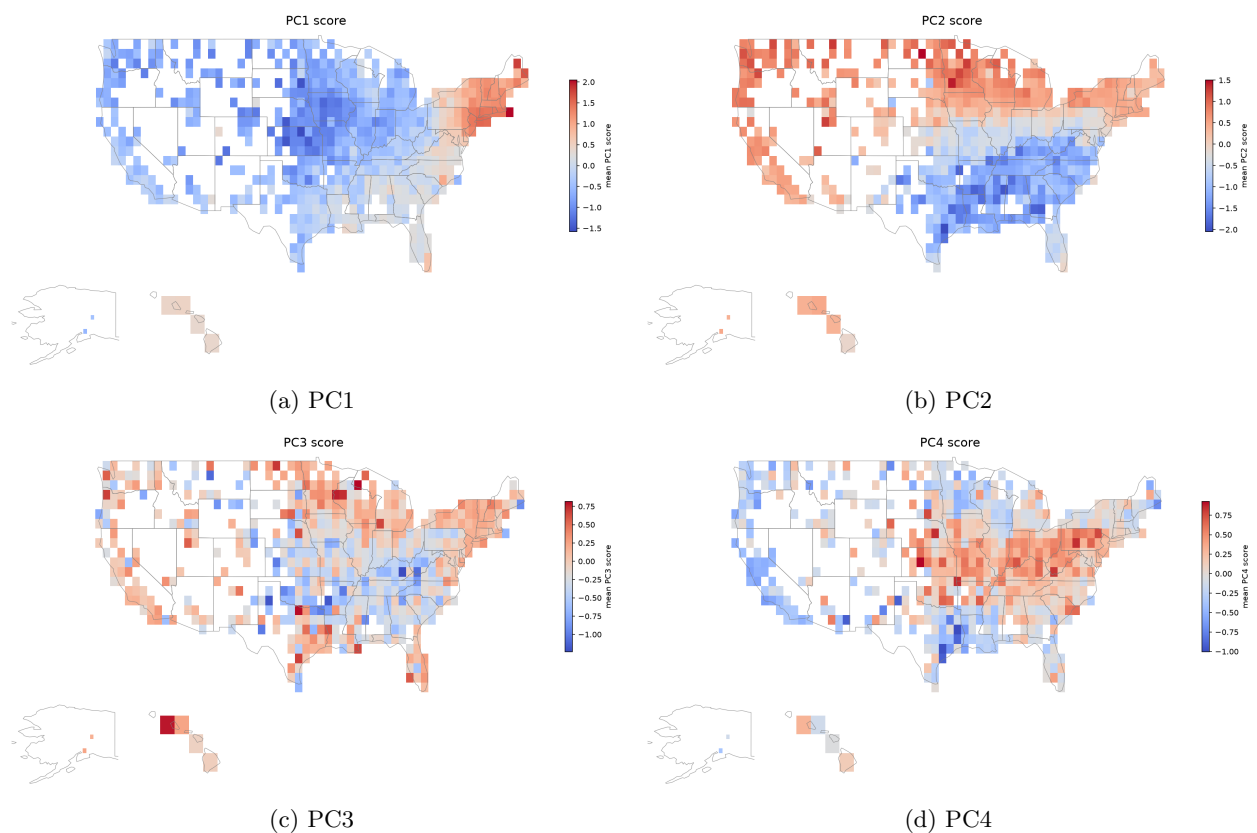


Figure 3: Mean PC score of respondents in each $1^\circ \times 1^\circ$. Bins with fewer than 5 respondents were dropped. Each map has its own color scale, running from its lowest to its highest bin mean.

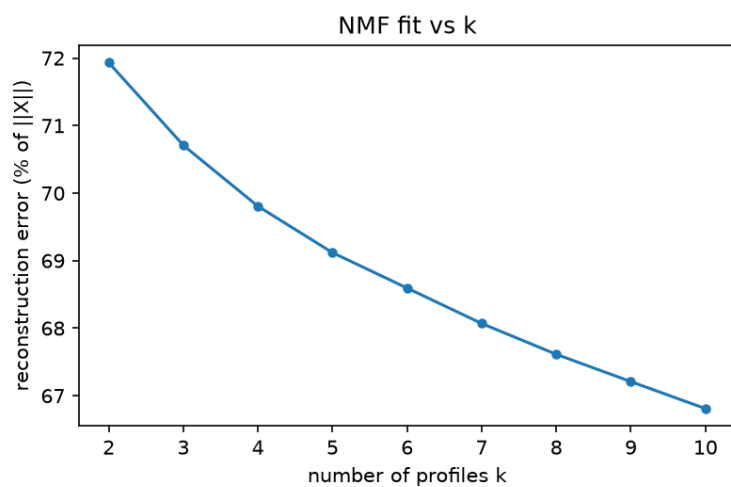


Figure 4: Reconstruction error by each NMF profile added.



Figure 5: NMF with $k = 4$ profiles. Each respondent's weights are to percentages summing to 100; maps show the mean percentage on each profile in each $1^\circ \times 1^\circ$ bin. Bins with fewer than 5 responses were dropped. Each map has its own color scale.

4 Clustering

Next, I wanted to see whether we could find data-driven clusters that would align with the geographical boundaries. In order to preserve more of the signal with more variance explained, I decided to run the clustering analyses on the first 10 PC scores.

First I did k-means clustering. In order to choose the K , I looked at the silhouette score using 5-fold cross-validation. I first split the data into 5 folds, then fit k-means on 4 folds. Then I assigned the held-out fold to the nearest center of the cluster and computed the silhouette score. This was repeated and averaged across the folds. $K=3$ scored the highest (figure 6a), so I used $K=3$ for k-means clustering. Note that most people are only slightly closer to their own cluster than to the second nearest (figure 6b), suggesting that these cluster identities are not very clear cut.

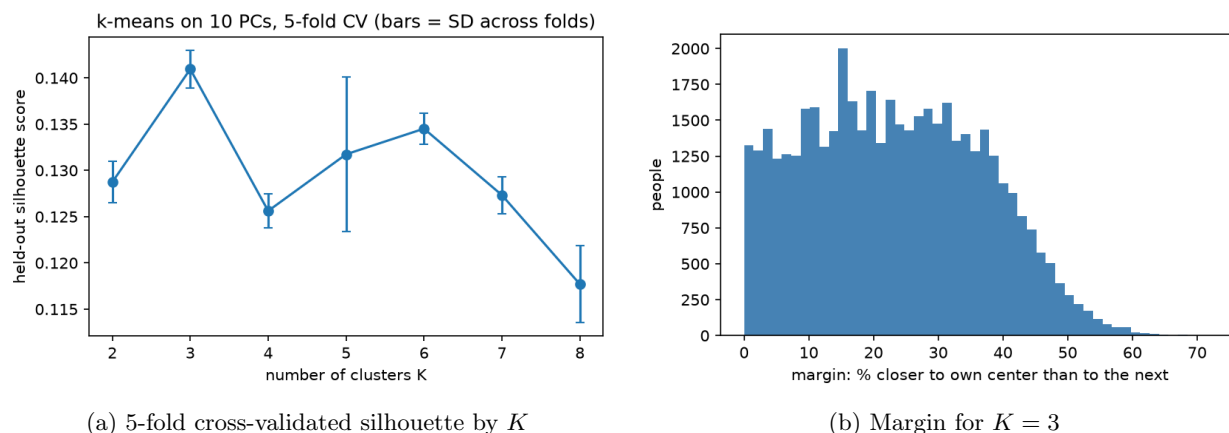


Figure 6: (a) Mean silhouette of held-out respondents, assigned to the nearest center of k-means fit on the other four folds. (Bars: SD across folds) (b) How much closer each respondent is to their own center than to the second nearest.

The clusters map (figure 7) again shows Northeast, Midwest+West, and South regions. It also shows that there are gradual transitions between these regions, as the opacity of the bins at the boundary are lower.

Next, I ran hierarchical clustering on a random subset of 5000 people, because the full pairwise distance matrix is too large. At each step, the two groups that add the least within-cluster variance are merged (figure 8a). If we cut the clusters at 3, we can see that it shows overlapping regions as the k-means clusters (figure 8b). The Northeast cluster splits off first, then the rest is split more between the North and South regions. The boundary between the Midwest and South looks to be shifted a bit more towards the North compared to the k-means map (figure 7). Overall, the two methods show the same core regions, with variations on where the transition zone lies. This further indicates that the dialect differences are better characterized as a continuum, as the specific algorithm used can place the boundaries between the clustered regions slightly differently.

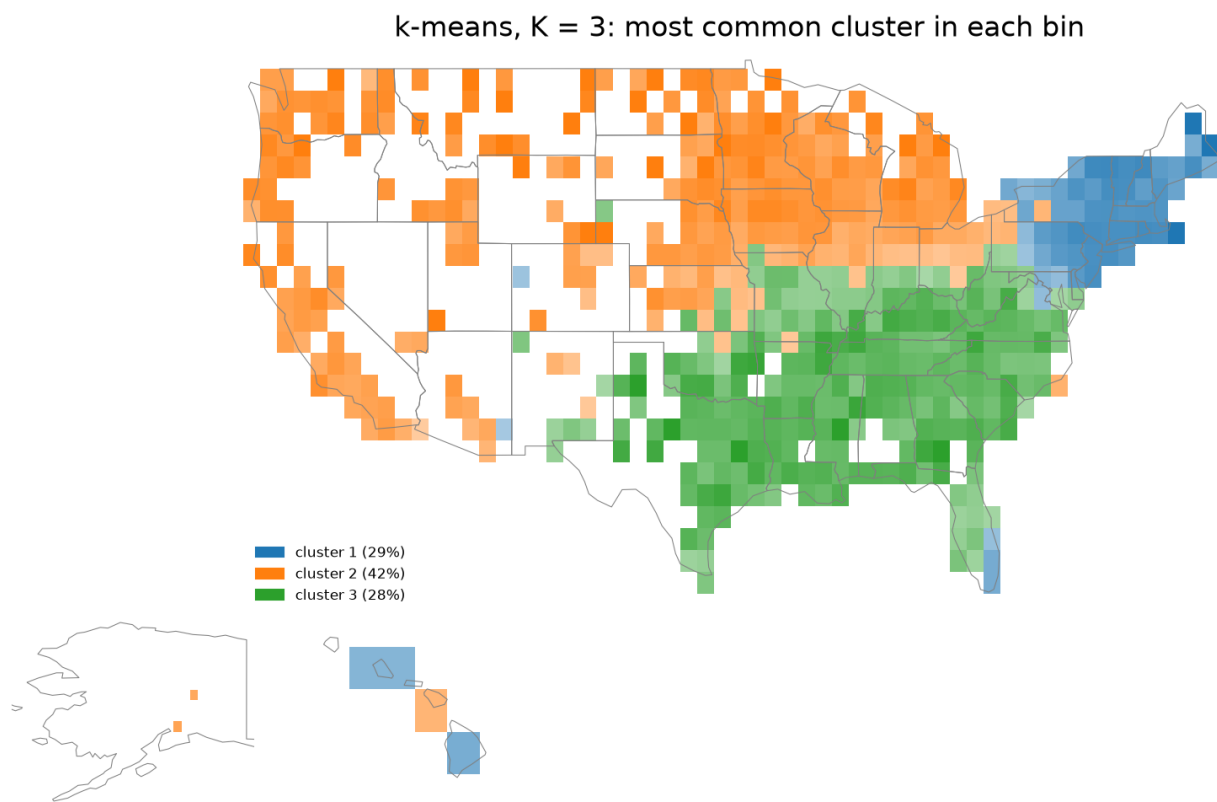
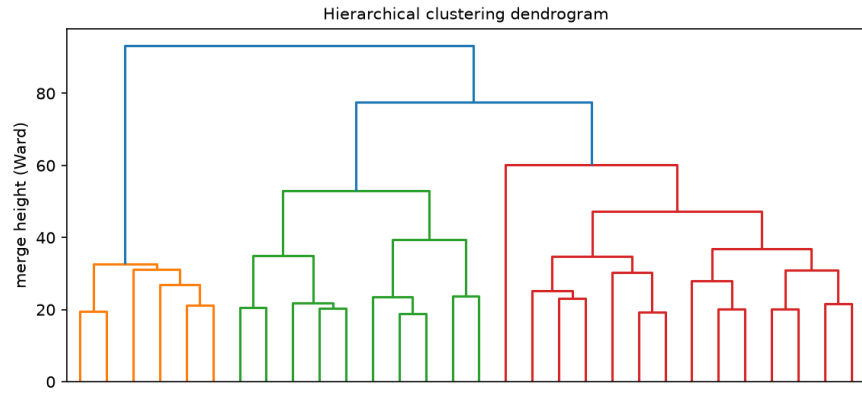
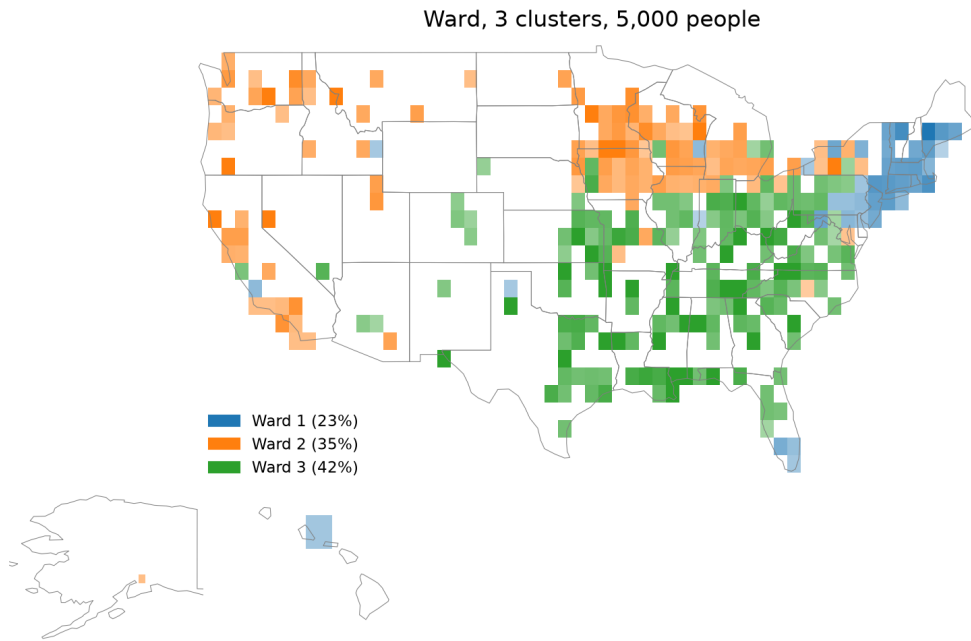


Figure 7: Most common k-means cluster in each bin



(a) Dendrogram (top 30 merges)



(b) Most common Ward cluster in each bin, cut at 3 clusters

Figure 8: Ward hierarchical clustering of a random 5,000 respondents on the same 10 PC scores. In (b), bins with fewer than 3 sampled respondents are omitted; opacity shows the percentage of the bin in its most common cluster.

5 Stability of findings to perturbation

First, I tried to see if the k-means results would give the same clusters with different starting points (random seeds). I tried running 20 different k-means runs, each with a different random seed set. Second, I took random subsamples of 80% of the data and reran PCA and k-means. I also repeated this 20 times. Both attempts were then compared with the original clustering using ARI. This reflects how pairs of participants were assigned into same or different clusters. The ARI was 0.933–1.000 for the random starts and 0.976–0.993 for the subsamples. This suggests that there were not widely different clusters that people were assigned to across different iterations of perturbation to the pipeline as well as the data.

6 Conclusion

I think this is an interesting dataset as a way to widely survey language use across the country, but I also think there are some limitations. Given the nature of an openly available online survey, the respondents would likely be younger than the average population. There could also be regional bias with easier access to the internet and online resources. Finally, language continues to change, so any decision made based on this single dataset may not be adequate at capturing future language use patterns. The clusters found throughout these analyses have been quite consistent across different methods and data perturbations, but they also consistently show that the pattern itself is a gradient rather than clear divisions of regions. A reality check of our region results could be collecting new data from each of these core and boundary regions to see if the same patterns are replicated. It would also be interesting to see if there are large mountains, rivers, etc. that overlap with the boundary areas that would have made interaction or spread of dialects more difficult.

The main results from this analysis suggests that while single features of dialect usage are weakly related to geographic factors, there are clear regional patterns that show up with multiple features combined. The main regions of Northeast, Midwest/West, and South show up consistently, with gradual transition boundaries between them.

7 Academic Honesty

7.1 Statement

The work in this report is my own. Academic honesty is important to me because ultimately what I want to gain from putting in the time and resources for this class is a better understanding of how I can use these statistical methods to pursue my own research questions.

7.2 LLM Usage

I used LLMs to help with formatting the figures and with overall grammar check and Latex formatting help with the report. The LLM highlighted areas for change and I made the changes myself.

7.3 Collaborators

None

8 Bibliography

Nerbonne, John and William Kretzschmar (2003). “Introducing Computational Techniques in Dialectometry”. In: *Computers and the Humanities* 37.3, pp. 245–255.